

1. Title: Searching the Crosslinking Mass Spectrometry Universe for New Protein-Protein Interactions

2. Contact Details:

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3. Abstract:

Crosslinking mass spectrometry (XL-MS) is a powerful structural proteomics method that can provide high-resolution structural information about complex mixtures of proteins in their native physiological context.^{1,2,3} The method works by chemically crosslinking two reactive amino acid residues that are close to one another in 3-D space, thereby freezing this spatial information into a covalent bond, and then retrieving this information by sequencing crosslinked peptides following enzymatic digest of the constituent proteins (like Hi-C,⁴ but for proteins). Initially developed in the 2000s to interrogate purified proteins, by the late-2010s, methods became available to interrogate protein-protein interactions (PPIs) on the proteome-scale and in the cellular environment.⁵ This project seeks to standardize, combine, and integrate existing publicly available crosslinking datasets from PRIDE to make these data more re-usable and useful for integrative and hybrid modeling. We will also create a meta-dataset to catalogue PPIs in human and other model organisms. Because XL-MS encodes information about both protein identity and interacting residues, these findings can be cross-validated to structural predictions of protein complexes using AlphaFold3, thereby providing stronger evidence for their existence compared to huMAP3.0, which predicts protein complexes without structural evidence. We will integrate these findings into the EBI Complex Portal to make it accessible to the life science community.

4. Introduction:

Every protein is connected.⁶ The physical interactions made by proteins oftentimes define their functions; hence, to understand the whole proteome, we must also have a complete compendium of protein complexes. Several proteomics approaches can be used to infer the existence of a pairwise protein-protein interaction (the fundamental unit of data used to build a model of a protein complex), such as affinity purification mass spectrometry (AP-MS), proximity labeling,⁷ co-fractionation mass spectrometry (CF-MS),⁸ and PAR-CL-MS (photoactivated RNA crosslinked coupled to mass spectrometry).⁹ BioPlex3.0 represents a particularly large-scale effort to catalogue human protein complexes,¹⁰ since the project's inception in 2015,¹¹ now 17,000 separate AP-MS experiments have been conducted by individually epitope-tagging nearly every human protein and then detecting which other proteins co-purify with it. huMAP3.0 combines BioPlex3.0 with data from CF-MS and proximity labeling and uses a machine learning model to cluster the PPI pairs into a global PPI network.¹² This network estimates there to be 15,000 complexes in the human proteome and that 14,000 proteins in the human proteome exist in some complex. These predictions were recently incorporated into the EBI Complex Portal, greatly expanding the 2,150 manually curated human protein complexes.^{13,14}

Whilst AP-MS, proximity labeling, and CF-MS are powerful methods, none of them provide structural information or evidence of which residues/interfaces are involved in a PPI. Moreover, AP-MS and CF-MS require that a complex survive lysis and remain intact *in vitro*, which means that they cannot generally capture proteins that are transiently associated with a complex.

Crosslinking mass spectrometry (XL-MS) is an orthogonal method for interrogating protein interactions that encodes a PPI by sequencing the two peptides derived from the two separate proteins covalently connected to each other through a chemical crosslinker (Figure 1A). Originally conceived as a structural method to be applied to purified proteins *in vitro*, advancements in crosslinker technology (i.e., MS-cleavable crosslinkers), mass spec instrumentation (i.e., high mass accuracy, sensitivity, and speed), and search software have enabled XL-MS to be applied to proteome-wide samples (Figure 1B).⁵ In other

words, a contact between any reactive residue X on any protein A to any reactive residue Y on any protein B can be sequenced.

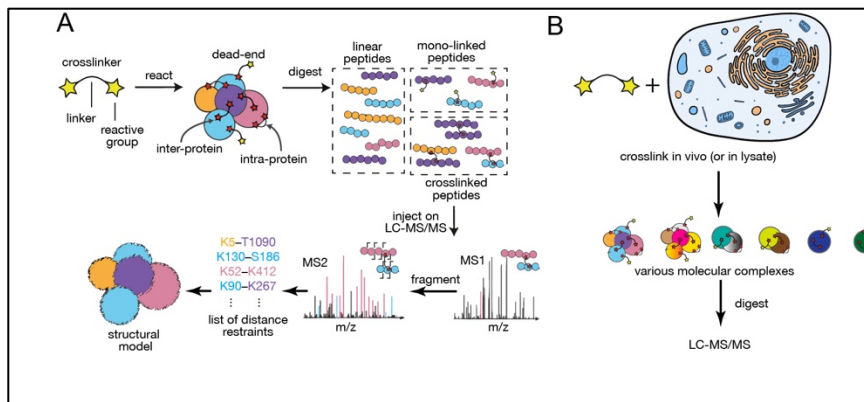


Figure 1. (A) Experimental scheme of crosslinking mass spectrometry. (B) Proteome-wide experiments involve performing crosslinking reactions on complex samples, like whole cells or extracts, which result in every protein complex getting crosslinked.

For a number of years, confidently identifying an inter-protein crosslink in a proteome-wide sample was considered very challenging, particularly because the search software did not have the capacity to estimate the FDR properly.^{15,16,17} This challenge has largely been overcome, and new algorithms (like Scout¹⁸) can perform these searches and validations quickly.

The XL-MS community has, in general, been amenable to making raw data available on PRIDE, one of the member repositories of the ProteomeXchange. Currently, there are ~1000 such datasets. However, it is not easy for these data to be reused or integrated for numerous reasons: (1) Over the years, many different search algorithms have been used, with various strengths and weaknesses (some are quite problematic); (2) Many experiments use customized fasta databases or search method parameters; (3) Until recently, there was not a unified format for reporting processed data (such as lists of residue-residue pairs), hence they were deposited to PRIDE as “partial crosslinking submissions”; (4) Sample metadata regarding tissue type, cell line, or disease conditions are often poorly annotated, posing significant challenges in linking the structure and complex to specific phenotypes or cellular molecular states.

This project has three proposed activities. **Firstly**, it aims to select large-scale XL-MS data in the PRIDE Archive, re-annotate them using SDRF (Sample and Data Relationship Format¹⁹) and reanalyze them using modern search algorithms, so that their findings can be more trustworthy and more readily reused by computational structural biologists. **Secondly**, it aims to mine through those data to (a) search for potentially new PPIs, and (b) provide additional validation to currently annotated PPIs. The second goal is motivated by the desire to provide more levels of validation to the large number of complexes that were put onto Complex Portal through huMAP 3.0. **A third goal**, which may not get finished during the project period, is to leverage information about PPIs implicating proteins of unknown function to begin proposing new functions to them.

5. Proposed Activities

5a. Reanalysis on XL-MS data on PRIDE. As discussed above, there are 1,143 datasets on PRIDE that mention crosslinking, though only a small number (23) are considered “complete” submissions. This is because full submissions require processed data to be included as .mzIdentML file, which many crosslinking search software did not do till recently (because it wasn’t supported yet).²⁰ Moreover, many datasets are limited to technical studies on a few protein structures, often lacking biological relevance. We will compile an extended list of biologically meaningful, large-scale XL datasets for protein complex and PPI analysis (a preliminary list is included in the proposal as part of the Datasets Table). For each legacy submission, we will reannotate the dataset in SDRF with the sample metadata. Using the raw spectra, a fasta file, and user-inputted description containing essential information (like which crosslinker was used), we will reperform the search and validation using Scout 1.5 and then put these results back onto PRIDE under new PXD accessions as Complete submission in standard file format mzIdentML and a link to the original submission will be provided. After the resubmission, the data will be available in the

novel PRIDE Crosslinking database (<https://www.ebi.ac.uk/pride/archive/crosslinking>). This activity will be facilitated through the EMBL-EBI API for fetching many large raw data files.

5b. Conglomerate Pairwise Residue Interactions to Validate Predicted Complexes. For this aim, we will likely focus on a smaller, curated, subset of the ~1,00 reprocessed XL-MS accessions across multiple model organisms. The criteria for inclusion will be: (i) proteome-wide experimental design (whether it be in lysate or in-cell); (ii) use of MS-cleavable crosslinker (i.e., DSSO); (iii) experiments were in a model organism (i.e., yeast, HEK cells); (iv) scale of the submission is above some cut-off (i.e., >10 raw files). The total set of inter-protein crosslinks from these projects will be pooled together. We will be able to perform further analyses for human PPIs and yeast PPIs. For the human PPIs, we will first ask, are the two proteins in question associated with a complex in huMAP3.0¹²? If so, we will perform an AlphaFold3²¹ prediction on that protein pair and test if it predicts the two residues in the crosslinked residue-pair are close in the modeled interaction. If so, this will be taken as a higher level of validation for that pairwise interaction in the complex. The respective entry in the Complex Portal will be amended so that PPI edges that have this further structural support will be annotated as such; moreover, these PPI edges will also be labeled to include the detected residue-pairs between the two proteins and references to the relevant PRIDE accessions. A flowchart illustrating this activity is provided in Figure 2.

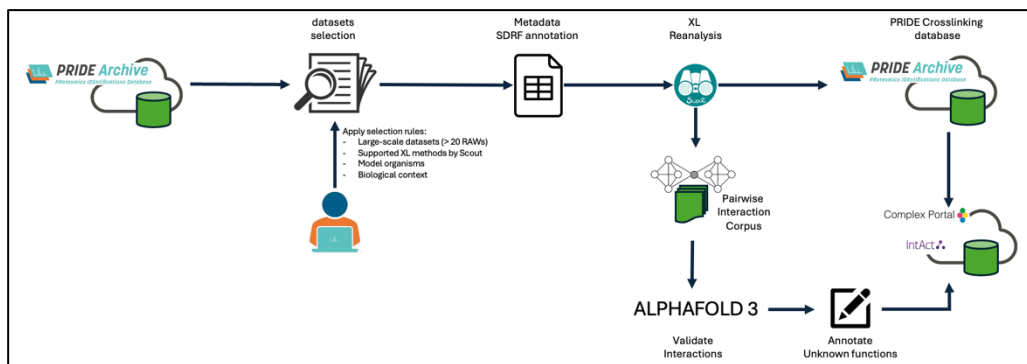


Figure 2. Flow chart illustrating activities a and b of this proposal.

This extra layer of validation for PPIs is a valuable contribution because 15,000 complexes in the human proteome does seem like an over-estimate, likely because AP-MS assays always have contaminants and bystander effects. XL-

MS data, whilst not as high throughput, do represent a more *direct* form of evidence for a physical interaction between two proteins.

For the PPIs that are not attested in huMAP3.0, we will also perform the AlphaFold3 prediction to see if the two residues are close in the modeled interaction. Those that pass this test will be added to Complex Portal under a provisional label that their evidence level is restricted to XL-MS.

A similar cross-validation campaign can be executed for yeast, as Matthias Mann's lab recently reported an interactome for yeast by performing 4,159 AP-MS (PRIDE PXD031940) experiments on (nearly) every protein in yeast.²²

If these objectives are completed earlier than expected, an additional activity we can set for ourselves is to reanalyze all AP-MS data on PRIDE Archive for organisms other than humans and yeast with quantms,²³ and then build a PPI network and cluster into complexes for each organism using the huMAP3.0 algorithm. This would then generate a list of complexes for a few non-yeast non-human model organism (likely *Drosophila* and mouse) that can also be compared to XL-MS data, following the same pipeline described above and in Figure 2.

5c. Propose Functions for Unannotated Proteins. According to one estimate, 15% of human proteins that can be assigned to orthogroups with other animal genes have unknown function (a "knownness score" of zero).²³ This is still a substantially large number of proteins (15% of 20,000 is 3,000). Many of

these proteins likely do not contain catalytic activities themselves, since these are normally executed by globular catalytic domains that possess detectable levels of homology to known folds. However, a potentially common function for a protein of unknown function is to serve as a scaffold to bring other proteins together at a specific time and place. Hence, a great way to begin annotating these proteins is to see what PPIs they form.

We propose to, in a limited form, apply this idea to the combined XL-MS dataset. For each validated PPI, we will assess its knownness score from the unknown. If one partner is 0, we will check if it interacts with a protein of known function. Examples of this will be compiled to see what patterns or network structures emerge. These data will likely be too sparse to merit updating any database of record with the findings; nevertheless, we hope the results can be published in some form and useful to create new hypotheses.

6. Outcomes

6a. Reanalysis on XL-MS data on PRIDE. The outcome of this activity will be recreation of ~100 large-scale different PRIDE submissions, reanalyzed in a standardized way and generating output in a standardised way, increasing the number of complete PRIDE crosslinking datasets from 23 (small datasets) to ~100 (biologically relevant large-scale submissions). This will facilitate re-use by enabling more users to view the residue-pair data without needing to be a mass spec expert or review the publications to find the residue-pairs in a large supplementary table. Moreover, it will make these data compatible for submissions to PDB-IHM if they are utilized for hybrid modeling.²⁴

6b. Conglomerate Pairwise Residue Interactions to Validate Predicted Complexes. Complexes on Complex Portal that currently have low evidence levels will have their evidence levels increased if they are structurally consistent with crosslinks. Meanwhile new protein complexes will be proposed if identified crosslinks are consistent with a preliminary AI based structural prediction. We expect the efforts of the first two aims could be the subject of a high-impact publication.

6c. Propose Functions for Unannotated Proteins. If a few dozen proteins of unknown function (as described above) can be found with known interactors, we would use those examples as a starting point for future experiments and seek further (non-NCMS) funding.

7. Working Group Approach

The project requires deep familiarity with crosslinking mass spectrometry analysis, working with databases, structural biology, and data science. It would be rare to find all these expertise in a single research group. Dr. Fried's team has a track-record on the structural biology and crosslinking mass spec side, but the EBI manages and runs PRIDE and Complex Portal, so Dr. Perez's expertise and support are essential for automating the processes of retrieving, analyzing, and redepositing data onto these databases in a consistent high-throughput manner. Moreover, the primary work of activity **a** will involve a large amount of manual curation, and hence is not suitable for a graduate student or post-doc project; it is better accomplished by a trained data scientist.

8. Rationale for NCMS Support

Even though it is very rich with information, the complexities of XL-MS data processing and the lack of standardization in processed data output prevent it from being re-used and integrated. XL-MS data speak to both protein structure and interaction, and unlike other PPI detection methods, it is unique in its ability to test predictions about protein complex structures, now broadly available through AlphaFold3, which has also recently been made open source.²⁵ Moreover, these modeling efforts are timely, as the RCSB transitions the PDB-dev to PDB-IHM (integrative and hybrid modeling) to highlight the greater importance of hybrid methods in structural biology.

9. Requested Resources

Activity **a** will require 6 months of a data scientist's FTE. This researcher will work closely with Dr. Perez and the EBI team. Hence, whilst they will be based at Penn State/NCEMS, we expect that they will benefit greatly from spending some time at EMBL-EBI in the United Kingdom for face-to-face interaction and cross-training, so some money for travel should be budgeted (\$10k). To estimate the compute costs for data reanalysis, we offer the following approximation: (100 crosslinking PXDs)×(200 raw files/PXD on average)×(~1h Scout run/raw file), which is 20,000 h on a workstation that has 20 cores @ 5.0 GHz and 32 GB of memory. Activity **b** is in the realm of structural biology rather than data curation, and so the science can be done by a grad-student/post-doc in the Fried lab whose salary comes from another grant. This activity will be computationally expensive, as we estimate there will be ~10,000 AlphaFold3 calculations on pairs of proteins with average lengths of 500 residues.

10. Proposed Timeline

12-15 months total.

Month 0-6 (activity **a**), project will be primarily driven by an NCEMS data scientist, advised by Dr. Perez-Riverol.

Month 6-12 (activity **b**), project will be primarily driven by a Fried lab trainee, advised by Drs. Fried and Perez-Riverol.

Month 12-15 (activity **c**), project is harder to define, but may involve a Fried lab trainee getting support from an NCEMS data scientist (in particular, with accessing data from the Unknome database).

12. Datasets Table

See attached.

13. Participant Table

At the present stage, confirmed participants for this project are those listed in item 2. It is likely that a student/post-doc from Dr. Fried's lab will participate, but who this is will depend on which rotation students join the lab.

14. Short CV of Project Leaders

See attached.

11. References

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Data Sources Table

Project Name:		Searching the Crosslinking Mass Spectrometry Universe for New Protein-Protein Interactions	
Datasets we will use - All crosslinking data on PRIDE - huMAP3.0 - Complex Portal			
Source ID (Use to reference in proposal narrative)	Source of data (URL/doi or name of individual database or primary literature)	Specific datasets of interest	Are there practical constraints on acquisition/sharing of this data?
1	https://www.ebi.ac.uk/pride	All that include “crosslinking” or “cross-linking” in description; can be queried through API	No; publicly available.
2	https://humap3.proteincomplexes.org/	Bulk downloads available	No; publicly available.
3	https://www.ebi.ac.uk/complexportal/	All human and yeast complexes; can be queried through API	No; publicly available.

List of large-scale crosslinking mass spectrometry datasets in PRIDE Archive

Dataset Accession	Title	Organism	Number of MS runs
PXD021114	HspB8 controls hardening and molecular aging of FUS condensates by domain-specific protein interactions Part 1	Homo sapiens	517
PXD035844	Utilising crosslinking MS to explore protein structure and interactions	Homo sapiens	401
PXD003754	Global Membrane Protein Interactome Analysis using In vivo Crosslinking and MS-based Protein Correlation Profiling	Homo sapiens	396
PXD002929	Global Membrane Protein Interactome Analysis using In vivo Crosslinking and MS-based Protein Correlation Profiling	Homo sapiens	276
PXD014759	Optimized parameter settings enhance proteome-wide formation of crosslinks on low-abundant proteins	Saccharomyces cerevisiae	240
PXD051075	Conventional cell fixation improves performance of in situ crosslinking mass spectrometry while preserving cellular ultrastructure	Homo sapiens Escherichia coli	226
PXD017695	In-cell architecture of an active transcription-translation complex	Mycoplasma pneumoniae	207
PXD010222	Chemical crosslinking mass spectrometry reveals the conformational landscape of the activation helix of PPAR γ ; a model for ligand-dependent antagonism	Homo sapiens	150
PXD015751	Cellular interactome dynamics during paclitaxel treatment	Homo sapiens	245
PXD026244	Structural probing of Hsp26 activation and client binding by quantitative cross-linking mass spectrometry	Saccharomyces cerevisiae	413
PXD013928	Proteome-wide Cross-linking mass spectrometry	Homo sapiens	344
PXD001183	Proteomic characterisation of microtubule-dependent modulation of adhesion complexes	Homo sapiens	240
PXD015435	Mass spectrometry study of formaldehyde cross-linking of structured proteins	Homo sapiens	211
PXD015687	Multi-layered proteomic analyses decode compositional and functional effects of cancer mutations on kinase complexes	Homo sapiens	355

PXD030532	Cross-linking of the Endolysosomal System Reveals Potential Flotillin Structures and Cargo	Homo sapiens	180
PXD032132	Cross-link assisted spatial proteomics to map sub-organelle proteomes and membrane protein topology	Homo sapiens Mus musculus	178
PXD042173	Proteome-scale recombinant standard datasets for XL-MS	Homo sapiens Escherichia coli	178
PXD014673	Integrative Structural Characterization of the Human Canonical and Non-canonical Cop9 Signalosome Complexes	Homo sapiens	141
PXD053500	Kinetic principles of chemical cross-link formation for protein-protein interactions	Homo sapiens	144
PXD030619	Mass Spectrometry Uncovers Protein Interactions and Stoichiometry of Intermediates and Sub-complexes for SNARE Complex Assembly	Rattus norvegicus	136
PXD031345	Normal respiratory chain capacity and physiology in mice with severely reduced levels of mitochondrial respirasomes; Cross-linking of intact mitochondria	Mus musculus	132
PXD042173	Proteome-scale recombinant standard datasets for XL-MS	Homo sapiens	178
PXD015160	Cross-linking mass spectrometry of synapses purified from mouse brain	Mus musculus	129
PXD007673	Chemical cross-linking mass spectrometry for systems structural biology in mouse heart tissue	Mus musculus	65
PXD022675	Extracellular whole cell cross-linking of B cells (JY)	Homo sapiens	123
PXD018771	Proteome-wide cross-linking mass spectrometry of human K562 cells	Homo sapiens	122
PXD010796	Optimized cross-linking mass spectrometry for in situ interaction proteomics	Homo sapiens	120
PXD010317	Cross-linking mass spectrometry of synaptosome	Homo sapiens	114
PXD058191	Cellular interactome dynamics during heat stress	Homo sapiens	112
PXD023817	Reverse-phase liquid chromatography (RPLC) in Combination with Tip-Based an Alternative Approach for Cross-linked Peptides analysis	Homo sapiens	105
PXD031054	Quantitative proteomics and in-cell cross-linking reveal cellular reorganisation during early neuronal differentiation of SH-SY5Y cells	Homo sapiens	102
			6480

Stephen D. Fried, Ph.D.

Assistant Professor of Chemistry
Abbreviated CV

Education and Degrees

2009–2014	Stanford University; Stanford, CA, USA Dissertation: <i>On the Origins of Catalysis by Ketosteroid Isomerase</i> Advisor: S. G. BOXER	Ph.D. Chemistry
2005–2009	Massachusetts Institute of Technology; Cambridge, MA, USA Thesis: <i>Oxygen-Oxygen Bonds: Catalytic Pathways in Energy Storage</i> Advisors: A. TOKMAKOFF & D. G. NOCERA	S.B. Chemistry Physics

Appointments and Positions

	Johns Hopkins University; Baltimore, MD, USA	
2021– <i>present</i>	Assistant Professor, T. C. Jenkins Department of Biophysics <i>by courtesy</i>	
2020– <i>present</i>	Assistant Professor, Department of Biology <i>by courtesy</i>	
2019– <i>present</i>	Preceptor, Chemistry-Biology Interface Program (CBI)	
2018– <i>present</i>	Assistant Professor, Department of Chemistry	
	University of Cambridge; Cambridge, UK	
2014–2018	Junior Research Fellow (King's College)	
	MRC Laboratory of Molecular Biology; Cambridge, UK	
2014–2018	Visiting Researcher	

Teaching Experience

2021– <i>present</i>	Chemical Biology II (AS.030.620), Guest Lecturer	JHU
2020– <i>present</i>	Planets, Life, and the Universe (AS.360.339), Guest Lecturer	JHU
2019– <i>present</i>	Physical Chemistry I (AS.030.301), Instructor	JHU
2018– <i>present</i>	Molecular Synthetic Biology (AS.030.623 / AS.030.424), Instructor	JHU
2016–2018	Biochemistry and Molecular Biology, Part IB, Supervisor	Cambridge
2015–2016	Chemistry, Part IA, Supervisor	Cambridge

Select Funding and Grants since 2020

2024	NIH/NCATS R41-TR005278 <i>“Designer Pichia Strains that Enable Difficult-to-Express Biologics Production by Addressing Limitations in Secretory Protein Expression”</i> Co-investigator (<i>with</i> Shane Byrne)	\$105k
2023	Impetus Grant (Hevolution & Rosenkranz Foundations) <i>“Can the Proteome Grow Old?: A Structural Proteomics Approach”</i> Principal Investigator (<i>with</i> Jeff Savas)	\$150k
2023	Camille Dreyfus Teacher-Scholar Award TC-23-019 <i>“How to Fold Every Protein – A Structural Proteomics Approach”</i> Principal Investigator	\$100k
2023	Human Frontiers in Science Program, Research Grant RGEC27/2023 <i>“Exploration of the structure/function space of prebiotic to biological proteins”</i> Co-investigator (<i>with</i> Klara Hlouchova & Kosuke Fujishima & Stephen Freeland)	\$350k (3 years)
2022	Cottrell Scholarship <i>“Bringing New Life to Prebiotic Peptide Chemistry and the Physical Chemistry Curriculum”</i> Principal Investigator	\$100k (3 years)
2021	NSF CAREER Award (MCB Division) MCB-2045844 <i>“Leaving the Fold: Leveraging Mass Spectrometry to Shift the Paradigm on Protein Folding”</i> Principal Investigator	\$847k (5 years)
2020	NIH New Innovator Award (NIGMS) DP2-GM140926 <i>“Watching Proteins Fold (or misfold) with Mass Spectrometry”</i> Principal Investigator	\$2.3M (5 years)

Stephen D. Fried, Ph.D.

Assistant Professor of Chemistry

Abbreviated CV

Select Publications since 2018

-
- 2024
(42) Romero-Pérez, P. S.; Moran, H. M.; Horani, A.; Truong, A.; Manriquez-Sandoval, E.; Ramirez, J. F.; Martinez, A.; Gollub, E.; Hunter, K.; Lotthammer, J. M.; Emenecker, R.; Liu, H.; Iwasa, J. H.; Boothby, T. C.; Holchouse, A. S.; **Fried, S. D.**; Sukenik, S.
“Protein surface chemistry encodes an adaptive tolerance to desiccation”
bioRxiv **2024**. DOI: 10.1101/2024.07.28.604841v2
Nature Chemical Biology. Under Review.
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PRX Life **2024**, 2, 033011.
- (38) Manriquez-Sandoval, E.; Brewer, J.; Lule, G.; Lopez, S.; **Fried, S. D.***
“FLiPPR: A Processor for Limited Proteolysis (LiP) Mass Spectrometry Datasets Built on FragPipe”
J. Proteome Res. **2024**, 23, 7, 2332–2342.
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“Cognition Associated Structural Changes across the Neuroproteome in a Rodent Model of Aging are Linked to Reduced Refolding Capacity”
Science Advances. In Revision.
- 2023
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“Secretion-catalyzed assembly of protein biomaterials on a bacterial membrane surface”
Angew. Chemie. Int. Ed. **2023**, 62, 37, e202305178.
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“Intrinsically Disordered Regions Promote Protein Refoldability and Facilitate Retrieval from Biomolecular Condensates.”
bioRxiv **2023**. DOI: 10.1101/2023.06.25.546465
Nature **2024**. Under Revision.
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- (33) Vu, Q. V.; Sitarik, I.; Jiang, Y.; Yadav, D.; Shama, P.; **Fried, S. D.***; Li, M. S.; O’Brien, E. P.
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Science Advances **2024**. Under Review.
- (31) Faustino, A. M.; Sharma, P.; Yadav, D.; **Fried, S. D.***
“Proteome-Wide Photo-Crosslinking Enables Residue-Level Visualization of Protein Interaction Networks in vivo”
Anal. Chem. **2023**, 95, 10670–10685.
- 2022
(30) Jacinto, M. P.; **Fried, S. D.***; Greenberg, M. M.
“Intracellular Formation of a DNA Damage-Induced, Histone Post-Translational Modification Following Bleomycin Treatment”
J. Am. Chem. Soc. **2022**, 144, 7600–7605.
- (29) Manriquez-Sandoval, E.; **Fried, S. D.***
“DomainMapper: Accurate domain structure annotation including those with non-contiguous topologies”
Protein Science **2022**, 31, e4465.
- (28) Jiang, Y.; Neti, S. S.; Pradhan, P.; Sitarik, I.; To, P.; **Fried, S. D.***; Booker, S. J.; O’Brien, E. P.
“How synonymous mutations alter enzyme structure and function over long time scales”
Nature Chem. **2022**, 15, 308–318.
- (27) Nissley, D. A.; Jiang, Y.; Trovato, F.; Sitarik, I. M.; Narayan, K.; To, P.; **Fried, S. D.***; O’Brien, E. P.
“Universal Protein misfolding intermediates can bypass the proteostasis network and remain soluble and non-functional.”
Nature Comm. **2022**, 13, 3081.
- (26) To, P.; Xia, Y.; Lee, S. O.; Devlin, T.; Fleming, K. G.; **Fried, S. D.***
“A Proteome-Wide Map of Chaperone-Assisted Protein Refolding in a Cytosol-like Milieu”
Proc. Natl. Acad. Sci. USA **2022**, 119, e2210536119.
- (25) **Fried, S. D.***; Fujishima, K.; Makarov, M.; Cherepashuk, I.; Hlouchova, K.
“Peptides before and during the nucleotide world: An origins story emphasizing cooperation between proteins and nucleic acids.”
J. Roy. Soc. Interface. **2022**, 19 (187), 20210641.
- (24) Nodelman, I. M.; Das, S.; Faustino, A. M.; **Fried, S. D.***; Bowman, G. D.; Armache, J.-P.
“Nucleosome recognition and DNA distortion by the Chd1 remodeler in a nucleotide-free state.”
Nature Struct. Mol. Biol. **2022**, 29, 121–129.

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Assistant Professor of Chemistry

Abbreviated CV

- 2021 To, P.; Whitehead, H.; Tarbox, H. E.; **Fried, S. D.***
(23) “Non-Refoldability is Pervasive Across the *E. coli* Proteome.”
J. Am. Chem. Soc. **2021**, *143*, 11435–11448.
- Lee, S. O.; **Fried, S. D.***
(22) “An Error Prone PCR Method for Short Amplicons.”
Anal. Biochem. **2021**, *628*, 114266.
- Lee, S. O.; Xie, Q.; **Fried, S. D.***
(21) “Optimized Loopable Translation as a Platform for the Synthesis of Repetitive Proteins.”
ACS Central Science **2021**, *7*, 1736–1750.
- 2020 Marx, D. C.; Plummer, A. M.; Faustino, A. M.; Roskopf, M. A.; Leblanc, M. J.; Devlin, T.; Lessen, H. J.;
(20) Majumdar, A.; Amann, B. T.; Flaming, P. J.; Kruegger, S.; **Fried, S. D.***; Fleming, K. G.
“*SurA* is a Cryptically Grooved Chaperone That Expands Unfolded Outer Membrane Proteins.”
Proc. Natl. Acad. Sci. USA **2020**, *117*, 28026–28035.
- Cervettini, D.; Tang, S.; **Fried, S. D.**; Willis, J. C. W.; Funke, L. F. H.; Chin, J. W.
(19) “Rapid and Scalable Discovery and Evolution of Orthogonal Aminoacyl tRNA Synthetase/tRNA pairs.”
Nature Biotech. **2020**, *38*, 989–999.
- Wu, Y.; **Fried, S. D.**; Boxer, S. G.
(18) “A Preorganized Electric Field Lead to Minimal Reorientation in the Catalytic Reaction of Ketosteroid Isomerase.”
J. Am. Chem. Soc. **2020**, *142*, 9993–9998.
- 2019 Saggu, M.; **Fried, S. D.**; Boxer, S. G.
(17) “Local and Global Electric Field Asymmetry in Photosynthetic Reaction Centers.”
J. Phys. Chem. B **2019**, *123*, 1527–1536.
- 2018 Schmied, W.; Tnimov, Z.; Uttamapinant, C.; Rae, C. D.; **Fried, S. D.**; Chin, J. W.
(16) “Controlling Orthogonal Ribosome Subunit Interactions Enables Evolution of New Function.”
Nature **2018**, *564*, 444–448.

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QUALIFICATIONS

2014	Ph.D. Biochemistry	University of Havana, Cuba
2006	B.Sc. Software Engineering	José Antonio Echeverría Technical Institute (ISPJAE), Havana, Cuba

APPOINTMENTS

2019-Present	Team Coordinator; EMBL-EBI
2017-2019	Project Lead; EMBL-EBI
2016-2017	Senior Software Engineer; EMBL-EBI
2014-2016	Bioinformatician; EMBL-EBI
2010-2016	Group Leader; Centre for Genetic Engineering and Biotechnology, Havana, Cuba
2006-2010	Bioinformatician; Centre for Genetic Engineering and Biotechnology, Havana, Cuba

PERSONAL STATEMENT

OTHER PROFESSIONAL ACTIVITIES & AWARDS

1. Associated Editor for four International Journal.
2. Reviews papers for more than 13 Journals.
3. Thomson/Clarivate Highly Cited Researcher since 2023 (<https://clarivate.com/tag/highly-cited-researchers/>)

SELECTED RECENT GRANTS

1. GRAPPA - Global comprehensive Atlas of Peptide and Protein Abundance (BB/T019670/1 - Co-Investigators)

SELECTED RECENT PUBLICATIONS

Overall, I have co-authored just over 124 journal articles (**H-index 40**)

(<https://scholar.google.co.uk/citations?user=WiaZQfkAAAAJ>). Selected publications:

1. **Perez-Riverol Y**, Bandla C, Kundu DJ, Kamatchinathan S, Bai J, Hewapathirana S, John NS, Prakash A, Walzer M, Wang S, Vizcaino JA. The PRIDE database at 20 years: 2025 update. **Nucleic Acids Res.** 2025 Jan 6;53(D1):D543-D553. doi: 10.1093/nar/gkaf1011. PMID: 39494541; PMCID: PMC11701690.
2. Dai C, Pfeuffer J, Wang H, Zheng P, Käll L, Sachsenberg T, Demichev V, Bai M, Kohlbacher O, **Perez-Riverol Y**. quantms: a cloud-based pipeline for quantitative proteomics enables the reanalysis of public proteomics data. **Nat Methods.** 2024 Sep;21(9):1603-1607. doi: 10.1038/s41592-024-02343-1. Epub 2024 Jul 4. PMID: 38965444; PMCID: PMC11399091.
3. Pfeuffer J, Bielow C, Wein S, Jeong K, Netz E, Walter A, Alka O, Nilse L, Colaianni PD, McCloskey D, Kim J, Rosenberger G, Bichmann L, Walzer M, Veit J, Boudaud B, Bernt M, Patikas N, Pilz M, Startek MP, Kutuzova S, Heumos L, Charkow J, Sing JC, Feroz A, Siraj A, Weissner H, Dijkstra TMH, **Perez-Riverol Y**, Röst H, Kohlbacher O, Sachsenberg T. OpenMS 3 enables reproducible analysis of large-scale mass spectrometry data. **Nat Methods.** 2024 Mar;21(3):365-367. doi: 10.1038/s41592-024-02197-7. PMID: 38366242.
4. Bai M, Deng J, Dai C, Pfeuffer J, Sachsenberg T, **Perez-Riverol Y**. LFQ-Based Peptide and Protein Intensity Differential Expression Analysis. **J Proteome Res.** 2023 Jun 2;22(6):2114-2123. PMCID: PMC10243145.
5. Deutsch EW, Bandeira N, **Perez-Riverol Y**, Sharma V, Carver JJ, Mendoza L, Kundu DJ, Wang S, Bandla C, Kamatchinathan S, Hewapathirana S, Pullman BS, Wertz J, Sun Z, Kawano S, Okuda S, Watanabe Y,

- MacLean B, MacCoss MJ, Zhu Y, Ishihama Y, Vizcaíno JA. The ProteomeXchange consortium at 10 years: 2023 update. **Nucleic Acids Res.** 2023 Jan 6;51(D1):D1539-D1548. PMCID: PMC9825490.
6. Umer HM, Audain E, Zhu Y, Pfeuffer J, Sachsenberg T, Lehtiö J, Branca RM, **Perez-Riverol Y**. Generation of ENSEMBL-based proteogenomics databases boosts the identification of non-canonical peptides. **Bioinformatics.** 2022 Feb 7;38(5):1470-1472. PMCID: PMC8825679.
 7. **Perez-Riverol Y**, Bai J, Bandla C, García-Seisdedos D, Hewapathirana S, Kamatchinathan S, Kundu DJ, Prakash A, Frericks-Zipper A, Eisenacher M, Walzer M, Wang S, Brazma A, Vizcaíno JA. The PRIDE database resources in 2022: a hub for mass spectrometry-based proteomics evidences. **Nucleic Acids Res.** 2022 Jan 7;50(D1):D543-D552. PMCID: PMC8728295.
 8. Dai C, Füllgrabe A, Pfeuffer J, Solovyeva EM, Deng J, Moreno P, Kamatchinathan S, Kundu DJ, .. **Perez-Riverol Y**. A proteomics sample metadata representation for multiomics integration and big data analysis. **Nat Commun.** 2021 Oct 6;12(1):5854. doi: 10.1038/s41467-021-26111-3. PMID: 34615866; PMCID: PMC8494749.
 9. Sinitcyn P, Hamzeiy H, Salinas Soto F, Itzhak D, McCarthy F, Wichmann C, Steger M, Ohmayer U, Distler U, Kaspar-Schoenefeld S, Prianichnikov N, Yilmaz Ş, Rudolph JD, Tenzer S, **Perez-Riverol Y**, Nagaraj N, Humphrey SJ, Cox J. MaxDIA enables library-based and library-free data-independent acquisition proteomics. **Nat Biotechnol.** 2021 Dec;39(12):1563-1573. Epub 2021 Jul 8. PMCID: PMC8668435.
 10. Deutsch EW, **Perez-Riverol Y**, et. al. Universal Spectrum Identifier for mass spectra. **Nat Methods.** 2021 Jul;18(7):768-770. Epub 2021 Jun 28. PMCID: PMC8405201.
 11. Jarnuczak AF, Najgebauer H, Barzine M, Kundu DJ, Ghavidel F, **Perez-Riverol Y**, Papatheodorou I, Brazma A, Vizcaíno JA. An integrated landscape of protein expression in human cancer. **Sci Data.** 2021 Apr 23;8(1):115. PMCID: PMC8065022.
 12. Moreno P, Huang N, Manning JR, Mohammed S, Solovyev A, Polanski K, Bacon W, Chazarra R, Talavera-López C, Doyle MA, Marnier G, Grüning B, Rasche H, George N, Fexova SK, Alibi M, Miao Z, **Perez-Riverol Y**, Haeussler M, Brazma A, Teichmann S, Meyer KB, Papatheodorou I. User-friendly, scalable tools and workflows for single-cell RNA-seq analysis. **Nat Methods.** 2021 Apr;18(4):327-328. PMCID: PMC8299072.
 13. Hulstaert N, Shofstahl J, Sachsenberg T, Walzer M, Barsnes H, Martens L, **Perez-Riverol Y**. ThermoRawFileParser: Modular, Scalable, and Cross-Platform RAW File Conversion. **J Proteome Res.** 2020 Jan 3;19(1):537-542. PMID: 31755270; PMCID: PMC7116465.
 14. **Perez-Riverol Y**, Zorin A, Dass G, Vu MT, Xu P, Glont M, Vizcaíno JA, Jarnuczak AF, Petryszak R, Ping P, Hermjakob H. (2019). Quantifying the impact of public omics data. **Nat Commun.** 10(1): 3512. PMCID: PMC6683138.
 15. **Perez-Riverol Y**, Csordas A, Bai J, et al. (2019). The PRIDE database and related tools and resources in 2019: improving support for quantification data. **Nucleic Acids Res.** 47(D1): D442-D450. PMCID: PMC6323896.
 16. da Veiga Leprevost F, Grüning BA, Alves Aflitos S, Röst HL, Uszkoreit J, Barsnes H, Vaudel M, Moreno P, Gatto L, Weber J, Bai M, Jimenez RC, Sachsenberg T, Pfeuffer J, Vera Alvarez R, Griss J, Nesvizhskii AI, **Perez-Riverol Y**. BioContainers: an open-source and community-driven framework for software standardization. **Bioinformatics.** 2017 Aug 15;33(16):2580-2582. doi: 10.1093/bioinformatics/btx192. PMID: 28379341; PMCID: PMC5870671.

PERSONAL INFORMATION

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German
14 JULY 1970

WORK EXPERIENCE

Dates (from – to)
Occupation, Employer

October 2014 - present
Head of Molecular Systems, European Bioinformatics Institute

Dates (from – to)
Occupation, Employer

August 2015 – December 2019
Director of Bioinformatics, National Center for Protein Sciences Beijing, China

Dates (from – to)
Occupation, Employer

January 2005 - present
Team Leader, European Bioinformatics Institute

Dates (from – to)
Occupation, Employer

September 1997 – December 2004
Database programmer, Sequence Database Group Coordinator, EBI

Dates (from – to)
Occupation, Employer

December 1996 – August 1997
Research Assistant, National Center for Biotechnology research (GBF), Braunschweig, Germany

EDUCATION AND TRAINING

Dates (from – to)
Organisation
Title of qualification
Level

October 1990 - October 1996
University of Bielefeld, Germany
Diplom (M.Sc.) Bioinformatics
Mit Auszeichnung

SKILLS AND COMPETENCES

Languages

German	English	French
Mother tongue	Excellent	Fluent

Technical skills
and competences

Programming languages: Java, Perl, SQL
Operating systems: Unix, Windows
Data modeling: UML, XML

MEMBERSHIPS AND POSITIONS

Swiss Institute of Bioinformatics STRING SAB member since 3/2022
EU MultiMot, SAB member, 2015-2018
Human Proteome Organisation (HUPO), founding member, council member 1/2013-12/2022; Secretary General 2021-2022.
ANR "Investissements d'Avenir", France, SAB member 2010-2019
HUPO Proteomics Standards Initiative (PSI), Chair/EC since 2006
British Society for Proteome Research, EC member 2004-2016
Center for Translational Molecular Medicine, Netherlands, SAC member, 2008 – 2013
Genoplante, France, SAC member, 2008-2010

AWARDS

Beijing "High-caliber Talent from Overseas" 2017
Thomson/Clarivate Highly Cited Researcher since 2014
(<https://clarivate.com/tag/highly-cited-researchers/>)
ERASMUS European student exchange scholarship, 1994/1995
<http://www.ebi.ac.uk/about/people/henning-hermjakob>

GROUP WEB SITE

Selected Publications (from 300, h index 98) (full list at <http://tinyurl.com/hhepub>)

1. Balu S, et al. Complex portal 2025: predicted human complexes and enhanced visualisation tools for the comparison of orthologous and paralogous complexes. *Nucleic Acids Res.* 2024 Nov;gkae1085.
2. Fischer SN, et al. hu.MAP3.0: Atlas of human protein complexes by integration of 25,000 proteomic experiments. *bioRxiv*; 2024. doi: 10.1101/2024.10.11.617930.
3. Gretnier A, et al. ReactomeGSA: new features to simplify public data reuse. *Bioinformatics.* 2024 Jun;40(6):btac338.
4. Milacic M, et al. The Reactome Pathway Knowledgebase 2024. *Nucleic Acids Res.* 2024 Jan;52(D1):D672-D678.
5. Tiwari K, Kananathan S, Roberts MG, Meyer JP, Sharif Shohan MU, Xavier A, Maire M, Zyoud A, Men J, Ng S, Nguyen TVN, Glont M, Hermjakob H, Malik-Sheriff RS. Reproducibility in systems biology modelling. *Mol Syst Biol.* 2021 Feb;17(2):e9982.
6. Keating SM, et al. SBML Level 3: an extensible format for the exchange and reuse of biological models. *Mol Syst Biol.* 2020 Aug;16(8):e9110.
7. Ostaszewski M, et al. COVID-19 Disease Map, building a computational repository of SARS-CoV-2 virus-host interaction mechanisms. *Sci Data.* 2020 May 5;7(1):136.
8. Perez-Riverol Y, et al. Quantifying the impact of public omics data. *Nat Commun.* 2019 Aug 5;10(1):3512.
9. IMEx Consortium Curators. Capturing variation impact on molecular interactions in the IMEx Consortium mutations data set. *Nat Commun.* 2019 Jan 2;10(1):10.
10. Perez-Riverol Y, et al. Discovering and linking public omics data sets using the Omics Discovery Index. *Nat Biotechnol.* 2017 May 9;35(5):406-409.
11. Griss J, et al. Recognizing millions of consistently unidentified spectra across hundreds of shotgun proteomics datasets. *Nat Methods.* 2016 Aug;13(8):651-656.
12. Vizcaino JA, et al. ProteomeXchange provides globally coordinated proteomics data submission and dissemination. *Nat Biotechnol.* 2014 Mar 10;32(3):223-6.
13. Griss J, Foster JM, Hermjakob H, Vizcaino JA. PRIDE Cluster: building a consensus of proteomics data. *Nat Methods.* 2013 Feb;10(2):95-6.
14. Cote R, et al. The PRoteomics IDentification (PRIDE) Converter 2 framework. *Mol Cell Proteomics.* 2012 Dec;11(12):1682-9.
15. Orchard S, et al. Protein interaction data curation: the International Molecular Exchange (IMEx) consortium. *Nat Methods.* 2012 Mar 27;9(4):345-350.
16. Wang R, et al. PRIDE Inspector: a tool to visualize and validate MS proteomics data. *Nat Biotechnol.* 2012 Feb 8;30(2):135-7. doi: 10.1038/nbt.2112.
17. Aranda B, et al. PSICQUIC and PSISCORE – Accessing and Scoring Molecular Interactions. *Nature Methods.* 2011 Jun 29;8(7):528-9.
18. Orchard S, et al. Minimum information about a bioactive entity (MIAPE). *Nat Rev Drug Discov.* 2011 Aug 31;10(9):661-9.
19. Salwinski L, et al. Recurated protein interaction datasets. *Nat Methods.* 2009 Dec;6(12):860-1.
20. Montecchi-Palazzi L, Kerrien S, Reisinger F, Aranda B, Jones AR, Martens L, Hermjakob H. The PSI semantic validator: a framework to check MIAPE compliance of proteomics data. *Proteomics.* 2009 Nov;9(22):5112-9.
21. Taylor, C. F., N. W. Paton, et al. The minimum information about a proteomics experiment (MIAPE). (2007) *Nat Biotechnol* 25(8): 887-93.
22. Kerrien S, et al. Broadening the Horizon - Level 2.5 of the HUPO-PSI Format for Molecular Interactions. *BMC Biol* (2007)5:44
23. Orchard S, et al. The Minimum Information required for reporting a Molecular Interaction Experiment (MIMIx). *Nat Biotechnol* (2007) 25(8): 894-8.
24. Hermjakob, H. et al. The HUPO PSI's molecular interaction format--a community standard for the representation of protein interaction data. *Nat Biotechnol* 22, 177-83 (2004).
25. Lander ES, et al. Initial sequencing and analysis of the human genome. *Nature.* 2001 Feb 15;409(6822):860-921.